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RESEARCH

Towards real-world multimodal propagation networks: Multimodal uncertainty graph contrastive learning for fake news detection

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Abstract
With the rapid evolution of social media platforms, the proliferation of fake news in various forms increasingly misleads readers. Existing multimodal and graph-based methods have successfully leveraged either cross-modal interactions or propagation networks to identify fake news. However, they struggle to generalize real-world situations, where news posts are rich in multiple modalities and widely disseminated across social networks. Furthermore, they overlooked the complementarity between multimodal information and propagation networks, as detection clues in multimodal news often emerge within comment propagation chains. To address these issues, we propose the Multimodal Uncertainty Graph Contrastive Learning (MUGCL) framework, the first attempt to integrate textual, visual, and propagation networks into a unified contrastive learning approach for fake news detection. Specifically, MUGCL constructs two parallel propagation chains, rooted in the textual and visual modalities of news, to capture propagation patterns of detection clues. Furthermore, we introduce an uncertainty-aware graph contrastive learning strategy to model collaborative interactions between different propagation chains and adaptively aggregate cross-modal uncertainties. Extensive experiments demonstrate that MUGCL significantly enhances fake news detection performance. Additionally, studies confirm that MUGCL effectively captures diverse real-world feedback to distinguish multimodal news, reduces dependence on news content, and improves generalization to complex social contexts.

Keywords Fake news detection · Graph contrastive learning · Multimodal learning · Propagation chains · Contrastive learning · Social media analysis

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1 Introduction

As social media platforms continue to expand, the rapid and widespread dissemination of false information has become an increasingly urgent issue. These platforms encourage individuals to create diverse forms of news content, including text, images, and videos, to attract greater attention. Meanwhile, news posts with huge propagation networks tend to spread widely, especially during periods of heightened international political tension and public health crises (Wu et al., 2023). For instance, during the 2016 United States presidential election, the top 20 fake news stories received significantly more shares and comments than authentic ones (Conroy et al., 2015a). Moreover, during the COVID-19 pandemic, a multitude of fabricated news stories in various formats emerged, resulting in a ninefold increase in the volume of news requiring verification (Simon et al., 2020). Therefore, detecting multiple forms of news on social media is crucial.

Existing research has focused on leveraging either news content or propagation networks for the task of fake news detection. Initially, news on social media primarily existed in text-based form and was often laden with malicious and manipulated content. Text-based methods, such as convolutional neural networks (CNN) (Yu et al., 2017), and pre-trained language models (Rao et al., 2021), have proven effective in detecting fake news. However, as social media platforms evolved, fake news creators began incorporating visual elements into news articles to make them more appealing and deceptive. Thus, multimodal methods (Galli et al., 2022; Singhal et al., 2020; Uppada et al., 2023) were developed to analyze the cross-modal relationships in multimodal news, demonstrating superior detection performance compared to text-based methods. Nevertheless, as fake news creators started imitating and plagiarizing content from authentic news sources, content-based detection methods became less effective (Ma et al., 2018). As a result, researchers have increasingly turned to graph-based approaches, leveraging propagation network features such as conversation graphs (Li et al., 2020; Ma et al., 2018; Song et al., 2021a, b) and propagation graphs (Min et al., 2022; Nguyen et al., 2020) to analyze the dissemination of fake news from a sociological perspective. Graph-based methods (Min et al., 2022; Nguyen et al., 2020) utilize their strong topology-aware modeling capabilities to capture explicit social context, thereby enhancing fake news detection.

While existing methods have made significant progress in detecting fake news on social media, the increasingly complex structure of multimodal content and the diversity of real-world user feedback pose new challenges. Prior approaches often overlook these intricacies, limiting their ability to capture the semantic inconsistencies that characterize deceptive content. One key challenge lies in the structural complexity of fake news, which now spans text, images, and their propagation networks. Beyond textual fabrication, malicious publishers deliberately introduce mismatches between visual and textual elements (Qi et al., 2021; Zeng et al., 2023a, b), manipulate comment sections with misleading information (Conroy et al., 2015a), and amplify false narratives through user propagation (Min et al., 2022; Nguyen et al., 2020). As illustrated in Fig. 1(a) (Wang et al., 2020), fake news manifests across text, images, and network structures, each playing a distinct role in information diffusion. Meanwhile, the diversity and noise in user feedback further complicate detection. As shown in Fig. 1(b), user comments often include conflicting responses–such as denial, affirmation, and sarcasm–that influence different modalities unequally. Existing methods mainly focus on text-based interaction chains (Li et al., 2020; Ma et al.,

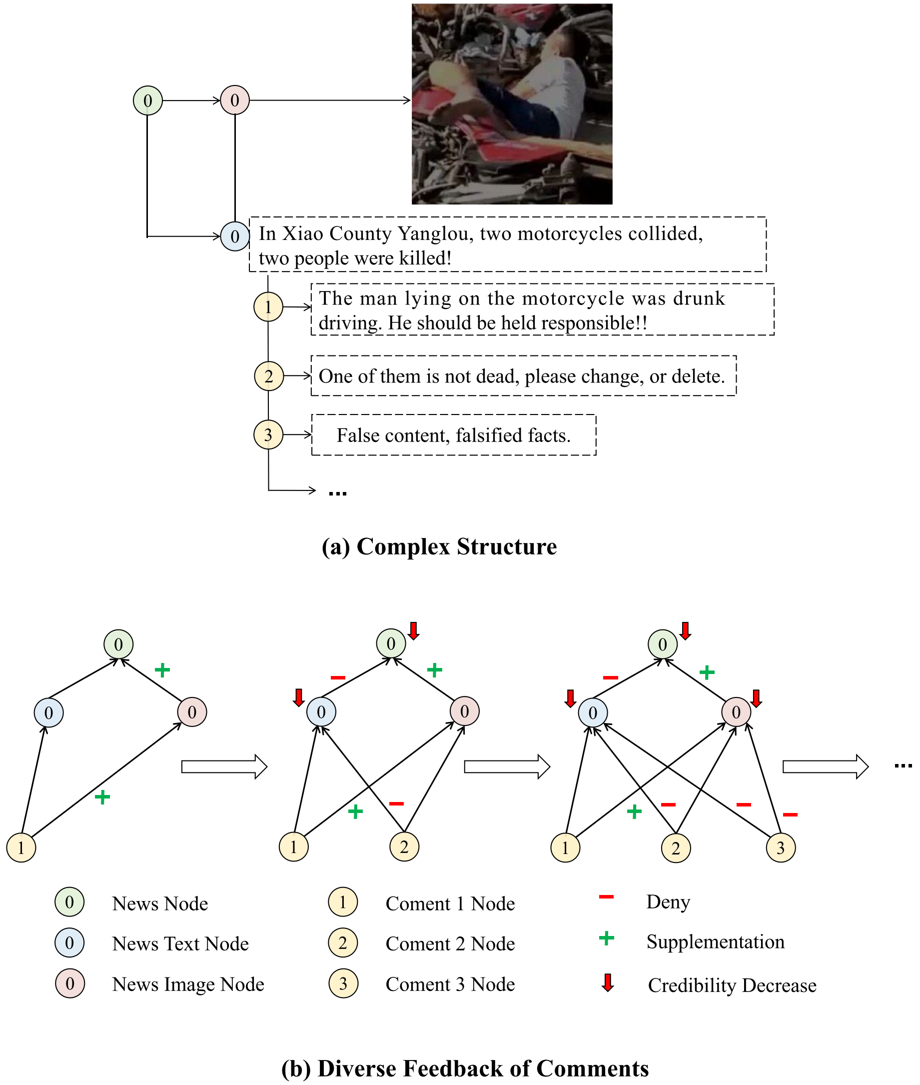

Fig. 1 Two examples of Structure Complexity and Feedback Diversity. (a) A example of complex structures of news. (b) The influence of diverse feedback on news in a propagation graph

2018; Song et al., 2021a), failing to account for such multimodal feedback dynamics. To address both challenges, we propose MUGCL, which introduces dual propagation chains rooted in text and image to independently model heterogeneous structures. Furthermore, we design a cross-modal uncertainty-aware contrastive learning module that quantifies semantic inconsistency between modalities and dynamically filters noisy signals. This unified framework enables MUGCL to robustly detect fake news in real-world, multimodal social contexts.

To address these challenges, we propose a Multimodal Uncertainty Graph Contrastive Learning (MUGCL) framework, which makes the first attempt to integrate textual, visual, and propagation networks into a unified contrastive learning framework for fake news identification. Specifically, we formulate the fake news detection task as a graph classification problem, where nodes represent news text, images, and comments, while edges capture their various interactions. This design enables the propagation of diverse feedback from comments within the propagation chains. More specifically, MUGCL constructs two propagation chains of comments, each rooted in a different news modality, to capture the propagation patterns of different modalities. Furthermore, we introduce a multimodal uncertainty graph contrastive learning strategy to model the collaborative interactions between different propagation chains and adaptively aggregate their cross-modal uncertainties. Consequently, MUGCL jointly leverages unimodal propagation patterns and multimodal propagation interactions to improve fake news detection. Extensive experimental results demonstrate that MUGCL significantly enhances the performance of fake news detection. Empirical studies further confirm that MUGCL effectively captures diverse real-world feedback to distinguish multimodal news, reduces reliance on news content, and enhances generalization for detecting fake news in complex social contexts.

The main contributions of this study are as follows: (1) We propose a novel paradigm of multimodal uncertainty-aware graph contrastive learning, making the first attempt to integrate textual, visual, and propagation networks into a unified contrastive learning framework for fake news detection. (2) Our MUGCL framework adaptively aggregates cross-modal uncertainty in multimodal propagation chains, effectively capturing both cross-modal consistency and inconsistency. (3) We evaluate MUGCL on two large-scale real-world datasets, demonstrating that our model outperforms state-of-the-art approaches. Furthermore, our findings confirm that MUGCL effectively leverages real-world feedback while reducing dependence on explicit news content.

2 Related work

2.1 Fake news detection

2.1.1 Content-based fake news detection

Early fake news on social media primarily existed in textual form, often containing malicious and manipulated content. Consequently, text-based models, such as convolutional neural networks (CNNs) (Yu et al., 2017), and pre-trained language models (Rao et al., 2021), were widely applied and demonstrated strong performance in detecting fake news. However, text-based methods struggle to detect fake news that integrates misleading visual elements. As multimodal learning evolved (Shi et al., 2025), fake news creators began incorporating visual elements into news articles, making them more engaging and deceptive. To address this limitation, multimodal detection approaches (Jin et al., 2017; Singhal et al., 2020; Wu et al., 2021) were introduced, leveraging cross-modal interactions to enhance detection performance. For instance, Singhal et al. (2022) explored cross-modal relationships and ambiguity (Chen et al., 2022) to improve fake news identification. Additionally, several studies have incorporated auxiliary features to refine multimodal learning, such as event-invariant features (Wang et al., 2018), noise-free features (Khattar et al., 2019), debiasing features (Zeng et al., 2024), video features (Zeng et al., 2025), LLM knowledge (Guo et al., 2025) and comment-based features (Wu et al., 2023), enabling more effective fake news detection.

2.1.2 Graph-based fake news detection

Beyond content analysis, researchers have explored graph-based approaches for fake news detection, leveraging the structural and social context of news dissemination (Huang et al., 2022; Qin et al., 2024). Studies have confirmed the effectiveness of graph-based features in capturing social context (Ma et al., 2018), leading to the development of models that represent fake news propagation as structured graphs. Specifically, researchers have constructed conversation graphs (Li et al., 2020; Rao et al., 2021; Song et al., 2021a) and propagation graphs (Min et al., 2022; Nguyen et al., 2020) to analyze how fake news spreads across social networks, providing sociological insights into its dissemination patterns. Graph-based methods (Nguyen et al., 2020) utilize their topology-aware modeling capabilities to incorporate explicit social context, significantly enhancing fake news detection. More recent advancements include heterogeneous graph neural networks (HGNNs) (Min et al., 2022) and hypergraph neural networks (HGNNs) (Jeong et al., 2022), which are designed to model the intricate relationships within social networks, further improving detection performance.

2.2 Graph contrastive learning

Graph contrastive learning (GCL) (Qin et al., 2024) is a crucial unsupervised learning approach that enables effective representation learning in graph-structured data. Existing GCL methods can be broadly categorized into structure-based and feature-based approaches. Structure-based methods primarily focus on reconstructing the local structure of a graph by applying perturbation techniques such as node dropping, edge perturbation, attribute masking, and subgraph extraction (You et al., 2020). Some studies have extended this approach by integrating global and local structure learning to enhance representation quality (Peng et al., 2020; Sun et al., 2019). Feature-based methods, on the other hand, aim to refine node representations by manipulating node features. For instance, Zhu et al. (2021) improved node feature reconstruction by introducing noise selectively to less important nodes. BGRL (Thakoor et al., 2021) employed a Siamese network to extract rich graph representations from two contrasting views, while MGCL (Liu et al., 2023) captured collaborative signals from multimodal interactions by leveraging modality-specific user preference features.

Table 1 provides a comparative analysis of our study and related work. Our approach is distinct in three key aspects: (1) The joint utilization of textual, visual, and graph features for enhanced multimodal representation learning. (2) The incorporation of propagation networks and multimodal information to refine graph-based representations. (3) The exploitation of multimodal interactions to facilitate more effective feature fusion and contrastive learning.

Table 1 A comparison of related studies

Model	TF	VF	CF	PS	MG	MI
EANN (Wang et al., 2018)	✓	✓				
MAVE (Khattar et al., 2019)	✓	✓				✓
Spofake+ (Singhal et al., 2020)	✓	✓				
BiGCN (Bian et al., 2020)	✓		✓	✓		
Stanker (Rao et al., 2021)	✓		✓			
CAFE (Chen et al., 2022)	✓	✓				✓
LIIMR (Singhal et al., 2022)	✓	✓				✓
MRHFR (Wu et al., 2023)	✓	✓	✓			✓
Ours	✓	✓	✓	✓	✓	✓

Column notations: Textual Feature (TF), Visual Feature (VF), Comment Feature (CF), propagation network (PS), Multimodal Graph (MG), Multimodal Interaction (MI)

3 Problem definition

Given a dataset of *m* news propagation graphs $C = \{c_1, c_2, \ldots, c_m\}$, where each c_i represents the propagation structure of the *i*-th news instance. Each c_i is defined as:

$$c_i = \{t_i, v_i, p_i^1, \ldots, p_i^{n_i}, G_i^t, G_i^v\} \tag{1}$$

where t_i and v_i are the root nodes corresponding to the textual and visual content of the news, respectively, and $\{p_i^1, \ldots, p_i^{n_i}\}$ represents the full set of comments associated with the post. Importantly, comments are not pre-assigned to specific modalities. Instead, we reuse the same set of comments to construct two parallel propagation graphs: the text-rooted graph G_i^t and the image-rooted graph G_i^v. This design assumes that some user responses may be influenced more by textual content and others by visual content, without requiring explicit annotation. The text-rooted propagation graph is defined as:

$$G_i^t = \{V_i^t, E_i^t\} \tag{2}$$

where $V_i^t = \{t_i, p_i^1, p_i^2, \ldots, p_i^{n_i}\}$ includes the news text and its associated comments. The edge set $E_i^t = \{e_i^{st} | s, t = 0, \ldots, n_i\}$ captures interaction patterns (e.g., replies, retweets). As shown in Fig. 2(a), node 0 represents t_i (text), and nodes 1 through n_i correspond to comments.

Similarly, the image-rooted propagation graph G_i^v shares the same comments but takes the image node v_i as the root:

$$V_i^v = \{v_i, p_i^1, p_i^2, \ldots, p_i^{n_i}\} \tag{3}$$

We use an adjacency matrix $A = \{a_i^{ts}\} \in \{0, 1\}^{(n_i+1)\times(n_i+1)}$ to describe the structure:

$$a_i^{ts} = \begin{cases} 1, & \text{if } (e_i^{ts} \in E_i^t \text{ or } E_i^v) \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

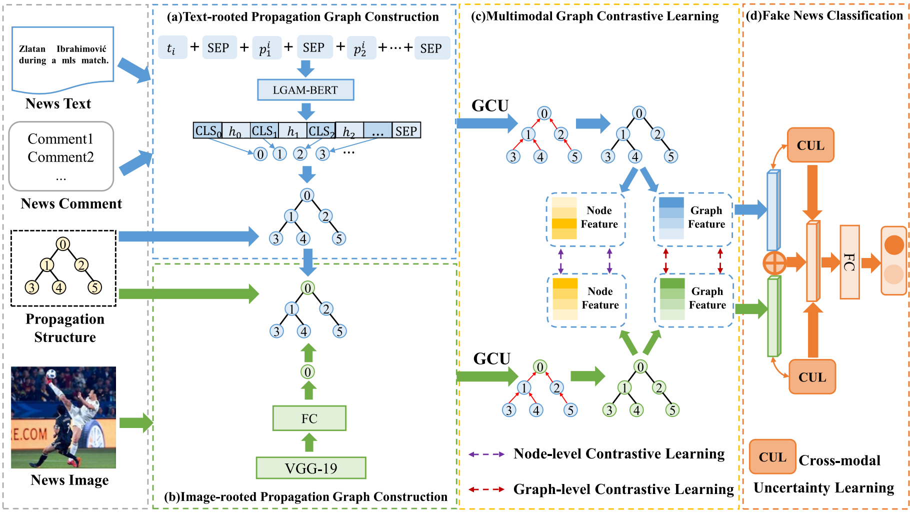

Fig. 2 The MUGCL is a detection framework that jointly model the textual, visual, and propagation networks into a contrastive learning framework. The textual, visual, and propagation networks are constructed into a text-rooted propagation graph and an image-rooted propagation graph, which learn complex feedback of unimodal propagation chain by the Graph Convolutional Unit (GCU). Finally, the multimodal uncertainty graph contrastive learning is applied to exploit multimodal propagation chain interactions and cross-modal uncertainty for fake news classification

We define fake news detection as a binary graph classification problem, where each news graph c_i is assigned a label $y_{c_i} \in \{0, 1\}$, with 0 denoting fake and 1 denoting real. The objective is to train a model $f(\cdot)$ that accurately predicts y_{c_i} for unseen c_i. Table 2 summarizes key notations.

Table 2 Notation summary

Notation	Definitions
c_i	The $i-$th news propagation graph
t_i	The news text of c_i
v_i	The news image of c_i
$p^i_{n_i}$	The n_i-th comment of c_i
G^t_i	The text-rooted propagation graph of c_i
G^v_i	The image-rooted propagation graph of c_i
X^t_i	The initial node features of G^t_i
X^v_i	The initial node features of G^v_i
N^t_i	The refined node features of G^t_i by the GCU
N^v_i	The refined node features of G^v_i by the GCU
s^t_i	The graph feature of G^t_i
s^v_i	The graph feature of G^v_i
y_{c_i}	The predicted label of c_i

4 Methodology

4.1 Propagation graph construction

4.1.1 Text-rooted propagation graph construction

To capture the diverse relationships between the news text and its responded comments, we construct a text-rooted propagation graph, denoted as $G_i^t = \{V_i^t, E_i^t\}$. As shown in Fig. 2(a), to extract the semantic correlations between the news text and comments, we select comments in chronological order and concatenate them using the special token [CLS], assigning position indices to the selected comments. The concatenation process is defined as follows:

$$D_i = [\text{CLS}] \oplus t_i \oplus p_1^i \oplus [\text{CLS}] \oplus p_2^i \oplus \cdots \oplus [\text{SEP}] \tag{5}$$

where $\oplus$ denotes the concatenation operation. LGAM-BERT (Rao et al., 2021) has demonstrated strong performance in encoding the propagation network of news and comments. Therefore, we encode D_i using LGAM-BERT:

$$X_i^t = \text{LGAM-BERT}(D_i) \tag{6}$$

The features of the news text and its associated comments learned by LGAM-BERT are represented as:

$$X_i^t = [x_0^t, x_1, \cdots, x_{n_i}] \in \mathbb{R}^{(n_i+1)\times d} \tag{7}$$

where x_0^t represents the feature vector of the news text, $\{x_1, \cdots, x_{n_i}\}$ represents the feature vectors of the comments, and d is the feature dimension. Finally, we initialize G_i^t with the encoded feature matrix X_i^t.

4.1.2 Image-rooted propagation graph construction

To capture the diverse relationships between the news image and its responded comments, we construct an image-rooted propagation graph, which can be denoted as $G_i^v = \{V_i^v, E_i^v\}$. We first employ the pre-trained VGG-19 (Simonyan & Zisserman, 2014) model to extract visual features of the news image from the spatial domain. We then extract the output V_g from the penultimate layer of VGG-19, which is pre-trained on the ImageNet dataset (Deng et al., 2009), and pass it through a fully connected layer $\sigma(\cdot)$ to project it into a d-dimensional space. The node feature of the news image, x_0^v, is computed as follows:

$$x_0^v = \sigma(W_v \cdot V_g) \tag{8}$$

where W_v is a learnable weight matrix. Next, we integrate the extracted image feature x_0^v with the comment features $\{x_1, x_2, \ldots, x_{n_i}\}$ from (7), resulting in the node features of the image-rooted propagation graph:

$$X_i^v = [x_0^v, x_1, \cdots, x_{n_i}] \in \mathbb{R}^{(n_i+1)\times d} \tag{9}$$

where d is the feature dimension, and X_i^v represents the initial node feature matrix of G_i^v.

4.2 Multimodal uncertainty graph contrastive learning

4.2.1 Graph convolutional unit

Feedback from the real world may provide supplementary, enhanced, or inconsistent information with respect to the news image (Qi et al., 2021). To effectively capture visual correlations from diverse user feedback in comments and refine the node features, we feed the node features $X_i^v = [x_0^v, x_1, \ldots, x_{n_i}] \in \mathbb{R}^{(n_i+1)\times d}$ into the Graph Convolutional Unit (GCU).

Given an image-rooted propagation graph $G_i^t = \{X_i^v, E_i^t\}$, where the node features are

$$X_i^v = [x_0^v, x_1, \ldots, x_{n_i}], \quad E_i^t = \{e_{st} \mid s, t = 0, \ldots, N\},$$

the GCU applies a graph convolution operation to aggregate neighborhood information.

$$h_{x_i}^l = \sigma\left(\sum_{j\in N(i)} W_1 \hat{A}_{ij} h_{x_i}^{(l-1)} \oplus {x_i}^{(l-1)} + b_1\right) \tag{10}$$

where $\oplus$ denotes the concatenation operation, and ${x_i}^{(l-1)}$ is the hidden feature computed by the $(l-1)$-th GCU layer. Then, we adopt a linear transformation $\sigma(\cdot)$ that maps from $\mathbb{R}^{2d}$ to $\mathbb{R}^d$, where both W_1 and b_1 are trainable parameters.

The GCU processes $X_i^v = [x_0^v, x_1, \ldots, x_{n_i}]$ in G_i^t and obtains the refined node features:

$$N_i^v = [h_{x_0}, h_{x_1}, \ldots, h_{x_{n_i}}] \tag{11}$$

Similarly, the GCU processes $X_i^t = [x_0, x_1, \ldots, x_{n_i}]$ and produces the refined node features:

$$N_i^t = [h_{x_0}, h_{x_1}, \ldots, h_{x_{n_i}}] \tag{12}$$

4.2.2 Node-level contrastive learning

MUGCL exploits local semantic interactions across multimodal propagation chains by aligning the representations of corresponding comment nodes from the text-rooted and image-rooted graphs. Specifically, we perform node-level contrastive learning on leaf nodes (i.e., comments) after graph encoding via the Graph Convolution Unit (GCU) and the Graph Gate Unit (GGU). For each pair of matched comment nodes $\{h_i^t, h_i^v\}$–extracted from G_i^t and G_i^v for the same news instance–we treat them as a positive pair, while mismatched comment nodes from different news samples serve as negative pairs.

Given a set of refined comment node feature pairs $\{h_i^t, h_i^v\}_{i=1}^m$, we define the node-level contrastive loss as follows (Gao et al., 2021):

$$\mathcal{L}_N = -\sum_{\{h_i^t, h_i^v\} \in \{N_i^t, N_i^v\}_{i=1}^m} \log \frac{e^{\mathrm{sim}(h_i^t, h_i^v)/\tau}}{\sum_{j=1}^m e^{\mathrm{sim}(h_i^t, h_j^v)/\tau}} \tag{13}$$

where τ is a temperature scaling parameter controlling the sharpness of the similarity distribution, and h_i^t, h_i^v denote the refined feature representations of corresponding comment nodes in G_i^t and G_i^v. The function $\mathrm{sim}(\cdot, \cdot)$ denotes cosine similarity. This contrastive objective guides the model to align semantically similar comment representations across modalities, enhancing robustness to modality inconsistency in fake news content.

4.2.3 Graph-level contrastive learning

Graph-level features of propagation graphs are obtained by aggregating G_i^t and G_i^v using a mean-pooling operation. The resulting graph-level representations for the text-rooted and image-rooted propagation graphs are denoted as s_i^t and s_i^v, respectively:

$$s_i^t = \mathrm{mean}(N_i^t) \tag{14}$$

$$s_i^v = \mathrm{mean}(N_i^v) \tag{15}$$

Graph-level contrastive learning is performed to align the representations of the text-rooted and image-rooted propagation graphs, ensuring more robust multimodal graph representations for fake news detection. It aims to learn more representative and intrinsic graph features by maximizing the agreement between s_i^t and s_i^v while contrasting them with other negative samples:

$$\mathcal{L}_G = -\sum_{i=1}^m \log \frac{\exp(s_i^t \cdot s_i^v)}{\exp(s_i^t \cdot s_i^v) + \exp(s_i^t \cdot s_i^-)} \tag{16}$$

where s_i^- represents a negative sample graph feature, typically selected from another propagation graph within the batch.

4.2.4 Cross-modal uncertainty learning strategy

To address the potential semantic misalignment between the text-rooted and image-rooted propagation graphs, we introduce a cross-modal uncertainty learning mechanism that quantifies the inconsistency between unimodal and fused multimodal representations. In real-world fake news scenarios, the same user comments may semantically align well with the textual content but conflict with the associated image (or vice versa). Directly fusing these propagation views without assessing their semantic agreement can lead to noisy or misleading representations.

To mitigate this, we first combine the unimodal graph representations s_i^t and s_i^v from G_i^t and G_i^v to form a unified multimodal representation s_i^m:

$$s_i^m = \sigma(W_t s_i^t \oplus W_v s_i^v + b_0) \tag{17}$$

where $\oplus$ denotes the concatenation operation, and W_t, W_v, and b_0 are trainable parameters of the fully connected layer $\sigma(\cdot)$, which maps from $\mathbb{R}^{2d}$ to $\mathbb{R}^d$.

To assess how well the unimodal semantics align with the fused representation, we compute the cosine similarity between each unimodal vector and the multimodal representation. For example, the textual-multimodal association is defined as:

$$M_i^t = \cos(s_i^t, s_i^m) \tag{18}$$

where $M_i^t \in \mathbb{R}^{d\times d}$, and d is the dimension of s_i^t and s_i^v. Similarly, we compute the visual-multimodal association M_i^v between s_i^v and s_i^m.

Finally, to quantify the overall semantic uncertainty between the propagation chains, we combine these two associations using a fully connected layer to produce the multimodal semantic uncertainty score M_i^m:

$$M_i^m = W_m(M_i^t \oplus M_i^v) + b_m \tag{19}$$

where W_m and b_m are learnable parameters, and $M_i^m \in \mathbb{R}^d$. This uncertainty-aware score is then used to modulate the contribution of multimodal information in downstream modules, helping the model reduce the influence of unreliable or misaligned propagation signals.

4.3 Fake news classification

Since different modality graphs share both consistency and uncertainty associations, we concatenate s_i^m and M_i^m for classification, which is defined as:

$$\hat{y}_i = \mathrm{softmax}(W_0(s_i^m \oplus M_i^m) + b_0) \tag{20}$$

where $\oplus$ denotes the concatenation operation, and W_0 and b_0 are trainable parameters of the fully connected layer, which maps from $\mathbb{R}^{2d}$ to $\mathbb{R}^d$. The predicted output $\hat{y}_i$ represents the classification score.

Since fake news detection is a binary classification task, we use cross-entropy as the loss function during training:

$$\mathcal{L}_C = -\sum_{i=1}^{m} [y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i)] \tag{21}$$

where Y is the set of actual labels. By considering both local and global interactions in multimodal propagation chains, the final optimization objective $\mathcal{L}_f$ is formulated as:

$$\mathcal{L}_f = \mathcal{L}_N + \alpha\mathcal{L}_G + \beta\mathcal{L}_C \tag{22}$$

where α and β are hyperparameters. This training strategy enables MUGCL to simultaneously capture unimodal propagation patterns and multimodal interactions, thereby improving fake news detection on social media.

5 Experiment

5.1 Dataset

To evaluate the effectiveness of the proposed MUGCL framework, we conduct experiments on two widely-used multimodal fake news detection datasets: Fakeddit and WeChat. These datasets provide large-scale, diverse news samples from different social media platforms, allowing for a comprehensive evaluation of the model's generalization across different contexts. Table 3 provides a summary of the dataset statistics.

Fakeddit is an English-language dataset collected from Reddit, a widely used news-sharing platform (Nakamura et al., 2020). It consists of news posts spanning 22 different categories, covering a broad range of topics such as politics, health, and entertainment. The dataset was constructed using a weak-supervision-based labeling approach, where fake and real news labels were automatically assigned based on metadata features such as user engagement (upvotes and downvotes) and interaction patterns. Each news instance in the dataset includes a textual description, an associated image, and user-generated comments. Due to its large scale and multimodal nature, Fakeddit has been widely used as a benchmark for evaluating fake news detection models.

In contrast, the WeChat dataset focuses on Chinese news articles collected from WeChat's Official Accounts between March 2018 and October 2018 (Wang et al., 2020). Unlike Fakeddit, which relies on weak supervision, WeChat's news samples were fact-checked and verified by domain experts, ensuring high annotation quality. Each news article includes a title, a full news body, an associated image, and a set of user comments. This dataset provides a valuable resource for studying fake news detection in a controlled Chinese-language social media environment. Compared to Fakeddit, WeChat exhibits a more structured and expert-verified dataset composition, making it an ideal benchmark for assessing the robustness of multimodal fake news detection models across different languages and platforms.

5.2 Comparison methods

To examine the performance of our proposed model, we compare it with several state-of-the-art fake news detection methods, including unimodal and multimodal approaches. These baselines encompass a variety of feature extraction and classification techniques, ranging from text-based and image-based models to advanced multimodal and graph-based frameworks. The comparison methods are summarized as follows:

Table 3 Statistics of datasets

Datasets	Fakeddit	Wechat
# of fake news	5,569	7,845
# of real news	4,299	2,742
# of images	9,868	10,587
# of comments	21,882	23,427

(1) BERT (Gao et al., 2021) is a text-based model that leverages a pre-trained Bidirectional Encoder Representations from Transformers (BERT) to extract deep contextualized representations from news text. The extracted features are then used to classify whether a news article is fake or real.

(2) VGG-19 (Simonyan & Zisserman, 2014) is an image-based model that applies a pre-trained VGG-19 convolutional neural network to extract visual features from news images. The extracted features are fed into a classifier to determine the authenticity of the news.

(3) EANN (Wang et al., 2018) is a multimodal framework that introduces the Event Adversarial Neural Network (EANN) to learn event-invariant features. By disentangling event-specific information, it aims to enhance the generalization capability of fake news detection.

(4) MAVE (Khattar et al., 2019) employs a Multimodal Variational Autoencoder (MAVE) to reconstruct multimodal features and generate noise-free representations for classification. This approach improves robustness by mitigating noisy or misleading signals in news data.

(5) Spotfake+ (Singhal et al., 2020) is a multimodal model that integrates a pre-trained XLNet for text feature extraction and a VGG-19 for image feature extraction. These features are fused to enhance the classification performance.

(6) BiGCN (Bian et al., 2020) is a graph-based model that captures propagation structures in fake news detection. It utilizes a bidirectional graph convolutional network (BiGCN) to explore features through both top-down and bottom-up news propagation networks.

(7) Stanker (Rao et al., 2021) applies a two-level-grained attention masked BERT framework for fake news detection. It captures fine-grained interactions in news content and user comments to enhance classification accuracy.

(8) MRHFR (Wu et al., 2023) introduces the Multi-Reading Habits Fusion Reasoning (MRHFR) network, which deeply fuses multimodal features and explores inconsistencies in multimodal information to improve fake news detection.

(9) T^3RD (Zhang et al., 2024) employs both global and local contrastive learning: the global component aims to learn invariant graph representations, while the local component captures invariant node representations to enhance fake news detection.

5.3 Implementation details

We split each dataset into a training set and a testing set using an 8:2 ratio. For the Fakeddit dataset, we adopt a binary classification setting for fake news detection. To construct an image-rooted propagation graph, we utilize a VGG-19 model pre-trained on the ImageNet dataset (Deng et al., 2009). In the node-level and graph-level contrastive learning stages, we set the temperature parameter to $\tau = 0.05$. For text representation, we set the maximum news text length to 100 tokens while employing the BERT-based uncased model (Devlin et al., 2019) for the Fakeddit dataset and the Chinese pre-trained BERT with Whole Word Masking (Cui et al., 2021) for the WeChat dataset. In our MUGCL framework, we set $\alpha = 1$ and $\beta = 1$ to balance the loss components. We evaluate model performance using Accuracy, F_1, Precision, Recall, and AUC as the evaluation metrics, with the highest values highlighted in bold in Table 4. For optimization, we adopt the Adam optimizer with a learning rate of 5e-5, a batch size of 16, and 50 training epochs.

Table 4 Experimental results of baselines and the proposed MUGCL

Method	Accuracy	F_1	Precision	Recall	AUC
Fakeddit Dataset					
BERT	0.7278± 0.02	0.6469± 0.01	0.6015± 0.01	0.6998± 0.01	0.7094± 0.02
VGG-19	0.7081± 0.01	0.5450± 0.01	0.6128± 0.01	0.4908± 0.01	0.6794± 0.01
EANN	0.7734± 0.01	0.7642± 0.02	0.7344± 0.01	0.7966± 0.02	0.7751± 0.02
MAVE	0.7745± 0.02	0.6037± 0.01	0.8071± 0.01	0.4822± 0.01	0.7864± 0.02
Spotfake+	0.7891± 0.02	0.7731± 0.02	0.7667± 0.02	0.7797± 0.01	0.7884± 0.02
BiGCN	0.8211± 0.02	0.7519± 0.01	0.7431± 0.01	0.7610± 0.01	0.8049± 0.02
T^3RD	0.8297 ± 0.03	0.7338 ± 0.02	0.8283± 0.02	0.6586 ± 0.03	0.7684 ± 0.02
Stanker	0.8155± 0.02	0.7451± 0.01	0.7338± 0.02	0.7568± 0.01	0.7984± 0.01
MRHFR	0.8383± 0.01	0.7758± 0.01	0.7667± 0.01	0.7852± 0.02	0.8230± 0.01
MUGCL	**0.8571**± 0.02	**0.8897**± 0.01	**0.8841**± 0.01	**0.8953**± 0.01	**0.8453**± 0.01
WeChat Dataset					
BERT	0.8810± 0.02	0.9212± 0.01	0.8836± 0.01	0.9621± 0.02	0.8772± 0.01
VGG-19	0.8573± 0.01	0.9007± 0.01	0.9007± 0.02	0.9007± 0.01	0.8942± 0.02
EANN	0.8734± 0.01	0.9170± 0.01	0.8722± 0.01	0.9667± 0.01	0.8752± 0.01
MAVE	0.8966± 0.01	0.9291± 0.01	0.9164± 0.02	0.9366± 0.01	0.8742± 0.02
Spotfake+	0.9131± 0.01	0.9394± 0.01	0.9487± 0.01	0.9302± 0.01	0.8874± 0.01
BiGCN	0.9140± 0.02	0.9408± 0.01	0.9140± 0.01	0.9445± 0.02	0.8944± 0.02
T^3RD	0.9102 ± 0.03	0.8211 ± 0.02	0.9140 ± 0.03	0.7453 ± 0.02	0.8558 ± 0.02
Stanker	0.8890± 0.01	0.9268± 0.01	0.8857± 0.01	0.9719± 0.02	0.8935± 0.01
MRHFR	0.9331± 0.01	0.9215± 0.02	0.9329± 0.02	0.8972± 0.01	0.8935± 0.01
MUGCL	**0.9461**± 0.02	**0.9633**± 0.01	**0.9487**± 0.01	**0.9785**± 0.01	**0.9436**± 0.02

For fair evaluation, we applied the same textual, and visual features of backbones and random seed for each comparison methods. Values are reported as mean ± standard deviation over five runs. **Bold**: best result

5.4 Performance comparison

To assess the effectiveness of our proposed model, we conduct comparative experiments against several state-of-the-art fake news detection methods. Among the nine evaluated detection models, MUGCL achieves the best performance. Specifically, our MUGCL outperforms the corresponding state-of-the-art methods by at least 1.88% and 1.30% in Accuracy on the Fakeddit and WeChat datasets, respectively. Fake news detection models that incorporate graph-based structures, such as BiGCN (Bian et al., 2020), and T^3RD (Zhang et al., 2024), generally achieve better performance compared to models that do not. This further highlights the importance of analyzing social network structures for robust fake news detection.

Multimodal methods, such as EANN (Wang et al., 2018), MAVE (Khattar et al., 2019), and Spotfake+ (Singhal et al., 2020), outperform unimodal approaches like BERT and VGG-19. These results highlight the importance of leveraging cross-modal interactions for improved fake news detection. With the increasing complexity of social networks, several methods have introduced real-world user comments to enhance their models, such as Stanker (Rao et al., 2021) and MRHFR (Wu et al., 2023), demonstrating enhanced modeling capabilities for news verification. Our MUGCL further integrates textual, visual, and propagation networks into a unified framework, enabling the effective utilization of multiple news structures on social media and ultimately improving detection performance.

Both MRHFR (Wu et al., 2023) and MUGCL leverage textual, visual, and comment-based information for fake news detection. However, MUGCL consistently outperforms MRHFR across both datasets, providing empirical validation that multimodal uncertainty graph contrastive learning, which leverages multimodal propagation networks, offers a more robust approach for fake news detection.

5.5 Ablation study

We conduct six ablation experiments to analyze the role of different components in MUGCL. Specifically, we create several internal models by removing certain inputs or modules, and the performance comparisons are shown in Table 5.

- **w/o Text**: We only use the news text, comments, and construct a text-rooted propagation graph. In the stage of graph contrastive learning, we change the positive sample pairs into text-text pairs.
- **w/o Vis**: We only use the news image, comments, and construct an image-rooted propagation graph. In the stage of graph contrastive learning, we change the positive sample pairs into image-image pairs.
- **w/o Comment**: We only use the news text and image. In the stage of graph contrastive learning, we only use features of the news text and image as the positive sample pairs.
- **w/o LGAM**: We replace the LGAM-BERT with a standard BERT (Devlin et al., 2019).
- **w/o GGU**: We replace the GGU with a standard GCN layer.
- **w/o GCL**: We remove the training stage of node-level and graph-level contrastive learning.

Table 5 Ablation experimental results of internal modules and the proposed MUGCL on two datasets

Method	Acc	F_1	Prec	Rec	AUC
Fakeddit					
w/o Text	0.8251±0.02	0.7505±0.01	0.7632±0.01	0.7382±0.01	0.8104±0.02
w/o Vis	0.7425±0.01	0.6350±0.02	0.6415±0.02	0.6287±0.02	0.7991±0.01
w/o Comment	0.8230±0.01	0.7832±0.01	0.7859±0.02	0.7805±0.01	0.8165±0.02
w/o LGAM	0.8348±0.01	0.8667±0.01	0.9014±0.01	0.8346±0.02	0.8348±0.01
w/o CUL	0.8180±0.01	0.7624±0.01	0.8193±0.01	0.7128±0.01	0.8019±0.01
w/o GCL	0.8459±0.01	0.7909±0.01	0.7998±0.01	0.7853±0.01	0.8316±0.01
MUGCL	**0.8571**±0.02	**0.8897**±0.01	**0.8841**±0.01	**0.8953**±0.01	**0.8453**±0.01
WeChat					
w/o Text	0.8947±0.01	0.9286±0.01	0.9109±0.01	0.9471±0.02	0.8781±0.01
w/o Vis	0.8843±0.01	0.9225±0.01	0.8945±0.01	0.9523±0.01	0.8722±0.02
w/o Comment	0.8866±0.02	0.9254±0.01	0.8831±0.01	0.9719±0.01	0.8917±0.02
w/o LGAM	0.9258±0.02	0.9486±0.01	0.9508±0.02	0.9464±0.02	0.9062±0.01
w/o CUL	0.9060±0.01	0.9370±0.01	0.9091±0.01	0.9667±0.02	0.9023±0.02
w/o GCL	0.9405±0.01	0.9591±0.01	0.9457±0.01	0.9634±0.01	0.9283±0.02
MUGCL	**0.9461**±0.02	**0.9633**±0.01	**0.9487**±0.01	**0.9785**±0.01	**0.9436**±0.02

Values are reported as mean ± standard deviation over five runs. **Bold**: best result

The results in Table 5 show that removing any structure from the news data leads to a performance drop, demonstrating the importance of leveraging multiple modalities. The text-rooted propagation graph has the most significant impact, as removing it results in a 6.18% decrease in accuracy. This confirms that while the propagation network is the most valuable, all three structures contribute to the overall performance of MUGCL in fake news detection.

In addition, the results also indicate that MUGCL outperforms all internal models, proving the effectiveness of cross-modal uncertainty learning in handling uncertainty across different modalities. To further explore the role of local and global interactions in diverse feedback, we remove graph contrastive learning and observe the impact. The results confirm that graph contrastive learning helps MUGCL learn these interactions more effectively, further enhancing its ability to detect fake news.

5.6 Sensitivity analysis

In this section, we have conducted a sensitivity analysis on the key hyperparameters α and β, which control the contributions of the intra-modal and inter-modal contrastive objectives, respectively. As shown in the updated Table 6, our model demonstrates stable performance across different values of α and β (1.0, 0.5, 0.1) on both the Fakeddit and WeChat datasets. This indicates that MUGCL is robust to the choice of these hyperparameters.

5.7 Visualization of MUGCL

To further demonstrate the effectiveness of MUGCL in feature learning, we visualize the distributions of the text-rooted propagation graph, the image-rooted propagation graph, and the final feature representations from the testing dataset in a two-dimensional space using t-SNE (van der Maaten & Hinton, 2008). As shown in Fig. 3, the number of false positive samples in the distributions is lower compared to before applying MUGCL. Overall, MUGCL improves the separability between positive and negative samples across both datasets. These results suggest that MUGCL effectively captures meaningful feature representations by integrating textual, visual, and propagation networks, thereby enhancing fake news detection performance.

Table 6 Sensitivity analysis of hyperparameters α and β on Fakeddit and WeChat datasets

Fakeddit	Accuracy	F_1	Precision	Recall
$\alpha = 1, \beta = 1$	0.8571 ± 0.02	0.8897 ± 0.01	0.8841 ± 0.01	0.8953 ± 0.02
$\alpha = 0.5, \beta = 0.5$	0.8530 ± 0.02	0.8894 ± 0.02	0.8624 ± 0.02	0.9181 ± 0.01
$\alpha = 0.1, \beta = 0.1$	0.8444 ± 0.01	0.8836 ± 0.01	0.8522 ± 0.01	0.9173 ± 0.02
WeChat	Accuracy	F_1	Precision	Recall
$\alpha = 1, \beta = 1$	0.9461 ± 0.02	0.9633 ± 0.01	0.9487 ± 0.01	0.9785 ± 0.02
$\alpha = 0.5, \beta = 0.5$	0.9339 ± 0.01	0.9545 ± 0.01	0.9508 ± 0.02	0.9582 ± 0.01
$\alpha = 0.1, \beta = 0.1$	0.9225 ± 0.01	0.9477 ± 0.02	0.9270 ± 0.02	0.9693 ± 0.02

Values are reported as mean ± standard deviation over five runs

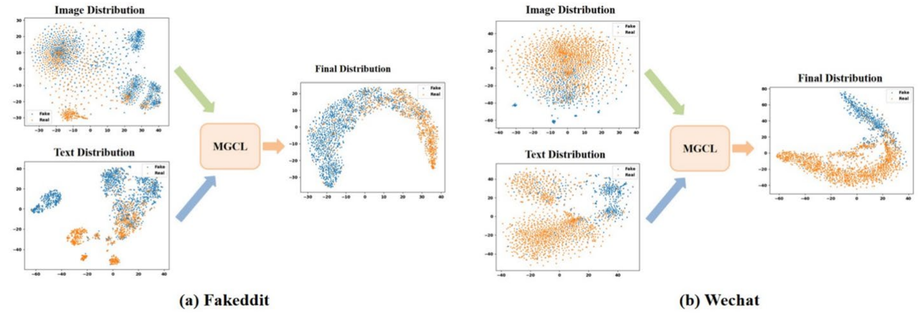

Fig. 3 Visualization of MUGCL on Fakeddit and Wechat datasets

5.8 Case study

In this section, we present a case study that visualizes the complex structures and diverse feedback of comments in propagation chains, offering insights into the effectiveness of MUGCL. We randomly selected two fake news samples and their corresponding multimodal propagation networks from the WeChat dataset, as shown in Fig. 4. This figure illustrates the hierarchical structure of news propagation, highlighting the relationships between textual, visual, and user-comment interactions.

To further analyze how MUGCL processes these structures, Fig. 5 provides a detailed visualization of the propagation networks and corresponding heatmaps, highlighting the key elements influencing MUGCL's decision-making process. In Fig. 5(c) and (g), the graph context demonstrates that MUGCL effectively captures the complex structural patterns of news propagation on social media, enhancing the generalization capability for fake news detection. Additionally, Fig. 5(d) and (h) present heatmaps that illustrate the varying importance of user comments in different propagation chains. The color intensity in the heatmaps indicates the attention MUGCL assigns to different comments, showing that it prioritizes more informative and reliable feedback while minimizing the influence of misleading comments. Furthermore, our observations suggest that MUGCL can leverage diverse user feedback to make accurate decisions in detecting fake news. By integrating textual, visual, and graph-based information, MUGCL enables the model to focus on more informative feedback from the real world, leading to more reliable predictions.

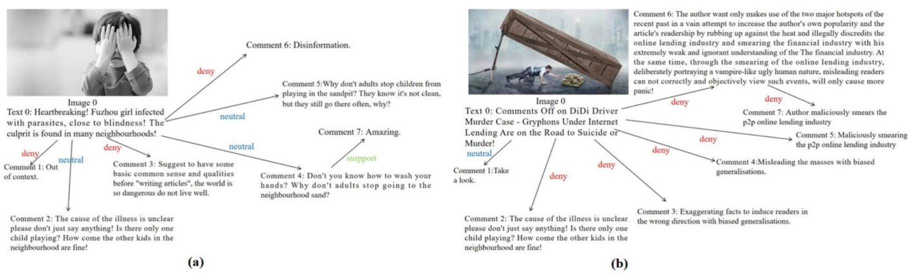

Fig. 4 The case study. Two fake news examples from Wechat dataset

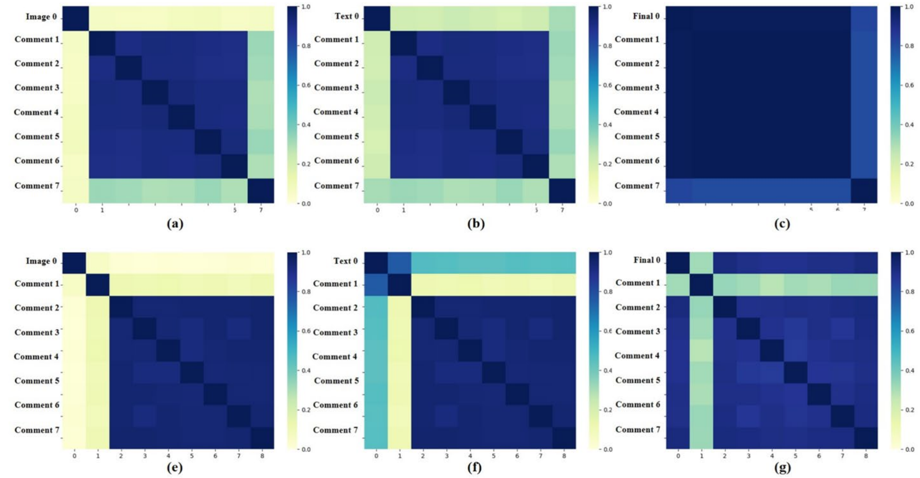

Fig. 5 Heat maps for two examples. Figure 5(a) and (b) illustrate the image-rooted and text-rooted propagation graphs corresponding to Fig. 4(a), while Fig. 5(c) shows the result after applying MUGCL. Figure 5(d) and (e) present the corresponding graphs for Figs. 4(b), and 5(f) shows the result after applying MUGCL to the same

These findings confirm that MUGCL reduces dependence on news content and improves its ability to detect fake news in complex social contexts.

6 Conclusion

In this study, we propose a multimodal uncertainty graph contrastive learning framework, which is the first attempt to integrate textual, visual, and propagation networks into a unified contrastive learning model for fake news detection. Extensive experimental results demonstrate that MUGCL improves detection performance. Moreover, this framework effectively captures diverse real-world feedback, providing a more comprehensive understanding of multimodal news dissemination. By leveraging uncertainty-aware cross-modal representations, MUGCL not only reduces reliance on explicit news content but also generalizes more effectively to complex and evolving social contexts where fake news manifests in diverse and deceptive forms.

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Declarations

Competing interests The authors declare no competing interests.

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